Abhijit Chowdhary

Cambridge, MA abhijit.chowdhary@tufts.edu abhijit-c.github.io Google Scholar

About Me

I consider myself a computational mathematician and my research interests lie in wielding math to design algorithms for real-world problems. I practice that in developing scalable algorithms to quantify, analyze, and control the uncertainty inherent in mathematical models. During my Ph.D., I focused on the uncertainty quantification (UQ) of large-scale inverse problems for physical phenomena governed by partial differential equations (PDEs). At the moment, I'm pursuing new directions in computational math, via problem structure-exploiting neural architectures, uncertainty quantification, randomized algorithms, among more.

Education

North Carolina State University, PhD in Applied Mathematics – Raleigh, NC — Aug 2020 – May 2025

- Recipient of the Siewert Fellowship

New York University, BA in Mathematics, Computer Science – New York, NY — May 2016 – May 2020

- Minor in Classics

Research Appointments

Postdoctoral Visiting Scholar, ICERM, Brown University – Providence, RI, USA — Jan 2026 – Apr 2026

Postdoctoral Researcher, Tufts University – Somerville, MA, USA — June 2025 – Present

- Working with Elizabeth Newman in structure-exploiting SciML and uncertainty quantification.
- Funded by Elizabeth Newman's 2025 AFOSR Young Investigator Program (YIP) Award (FA9550-26-1-B049).

Givens Intern, Argonne National Laboratory – Lemont, IL, USA — May 2024 – Aug 2024

Givens Intern, Argonne National Laboratory – Lemont, IL, USA — May 2023 – Aug 2023

- Became a core maintainer of the PyOED project and published a peer reviewed paper in TOMS.
- Published novel research in Robust Optimal Experimental Design, in review at SIAM CSE.
- Under the mentorship under Ahmed Attia.

Research Assistant, North Carolina State University – Raleigh, NC — June 2021 – Aug 2024

- Published novel research in scalable UQ for infinite-dimensional Bayesian inverse problems.
- Research Assistant for Alen Alexanderian via NSF Grant DMS-2111044.

Undergraduate Research Assistant, Ohio University – Athens, OH — May 2019 – Aug 2019

- Published novel research in the theory of pattern formation, for the 2D Swift-Hohenberg Equation.

Teaching

Teaching, Tufts University – Somerville, MA, USA

- Teaching Mathematical Modeling in Spring 2027.
- Teaching Mathematics of Data Science in Fall 2026.
- Led recitations for Linear Algebra in Fall 2025.

Teaching Assistant, North Carolina State University – Raleigh, NC

- Led recitations for Calculus I, II, and the honors variants for 5 semesters.

Publications

Exploiting Exact Conditionals Improves Conditioning: Provably Fast Mixing Time Bounds By Sampling from the Marginal Aug 2026
Abhijit Chowdhary, Federica Milinanni, Julianne Chung, Elizabeth Newman
https://doi.org/10.48550/arXiv.2608.27884 (arXiv preprint, submitted to SIAM JUQ)

Boost Like a (Var)Pro: Trust-Region Gradient Boosting via Variable Projection Mar 2026
Abhijit Chowdhary, Elizabeth Newman, Deepanshu Verma
https://doi.org/10.48550/arXiv.2603.23658 (arXiv preprint, submitted to JMLR)

Scalable Uncertainty Quantification for Infinite-Dimensional Bayesian Inverse Problems Feb 2025
Abhijit Chowdhary
www.lib.ncsu.edu/resolver/1840.20/45106 (PhD Thesis, NC State University)

Robust optimal design of large-scale Bayesian nonlinear inverse problems Sept 2024
Abhijit Chowdhary, Ahmed Attia, Alen Alexanderian
https://doi.org/10.48550/arXiv.2409.09137 (SIAM Journal on Scientific Computing)

PyOED: An Extensible Suite for Data Assimilation and Model-Constrained Optimal Design of Experiments June 2024
Abhijit Chowdhary, Shady E. Ahmed, Ahmed Attia
https://dl.acm.org/doi/10.1145/3653071 (ACM Transactions on Mathematical Software)

Sensitivity Analysis of the Information Gain in Infinite-Dimensional Bayesian Linear Inverse Problems May 2024
Abhijit Chowdhary, Shanyin Tong, Georg Stadler, Alen Alexanderian
https://doi.org/10.1615/Int.J.UncertaintyQuantification.2024051416 (IJUQ)

Weak Diffusive Stability of Roll Solutions at the Zigzag Boundary Oct 2023
Abhijit Chowdhary, Mason Haberle, William Ofori-atta, Qiliang Wu
https://doi.org/10.1137/24M1639683 (SIAM Journal of Mathematical Analysis)

Software

PyOED May 2023 – May 2025

- Core maintainer of PyOED, an open source research software for model-constrained optimal design.
- Designed to extend and integrate existing codes to rapidly test new developments.
- Used in multiple research groups worldwide.

Talks

Randomized Strategies for Variable Projection in Hierarchical Inverse Problems Nov 2026

- SIAM IS 2026, Salt Lake City, UT (Nov 2026).

VPBoost: Variable Projection for Gradient Boosting and VarProNets Dec 2025 – Present

- SIAM MDS 2026, Salt Lake City, UT (Nov 2026).
- WCCM–ECCOMAS 2026, Munich, Germany (Jul 2026).
- CMO Integrating Data- and Physics-Driven Methods for Decision Making under Uncertainty, Oaxaca, MX (June 2026)
- SIAM UQ 2026, Minneapolis, MN (Mar 2026).
- ICERM, Providence, RI (Feb 2026).
- CBMS Conference, University of Houston, Houston, TX (Dec 2025).

Robust Optimal Experimental Design for Large-Scale Bayesian Inverse Problems May 2023 – Mar 2025

- SIAM CSE 2025, Fort Worth, TX (Mar 2025).
- Applied Math Graduate Student Seminar, Raleigh, NC (Nov 2024).
- SIAM MDS 2024, Atlanta, GA (Oct 2024).
- SASSy Seminar, Argonne National Laboratory (May 2023).

PyOED for Bayesian Optimal Experimental Design Feb 2024 – Aug 2024

- SASSy Seminar, Argonne National Laboratory (Aug 2024).
- SIAM UQ 2024, Trieste, Italy (Feb 2024).

Sensitivity Analysis and Scalable Optimal Design for Bayesian Inference Apr 2022 – Aug 2023

- Applied Math Graduate Student Seminar, Raleigh, NC (Aug 2023).
- SIAM CSE 2023, Amsterdam, Netherlands (Mar 2023).
- Applied Math Graduate Student Seminar, Raleigh, NC (Apr and Sept 2022).

Service

Group Leader, VERSEIM-REU June 2026 – Aug 2026

- Leading a research group of undergraduates alongside Elizabeth Newman.

Co-Organizer, Postdoctoral and Graduate Student Seminar, ICERM Jan 2026 – Apr 2026

Lead Organizer, Applied Math Graduate Student Seminar (AMGSS), NCSU Jan 2022 – May 2025

Peer-Review for the following journals.

- Journal of Open Source Software (JOSS)
- SIAM Journal on Matrix Analysis and Applications (SIMAX)
- SIAM Journal on Uncertainty Quantification (JUQ)

Skills

Programming: Proficient in Python, Julia, C, C++, MATLAB, and Shell Scripting

Technologies: Generally familiar with the GPU programming interfaces, the machine learning stacks, Finite element software, high-performance linear algebra, and Linux.

Mathematics: Specialized in scientific computing, uncertainty quantification, and large-scale optimization. Theoretical foundations in the PDE, probability, and functional analysis.

Languages: English (native), Latin, Hindi